

Lesson 13 Demo 03

Implementing Intersite Connectivity

Objective: To implement a lab environment with different networks having full connectivity between them and verify its functionality

Tools required: Azure account with administrator access

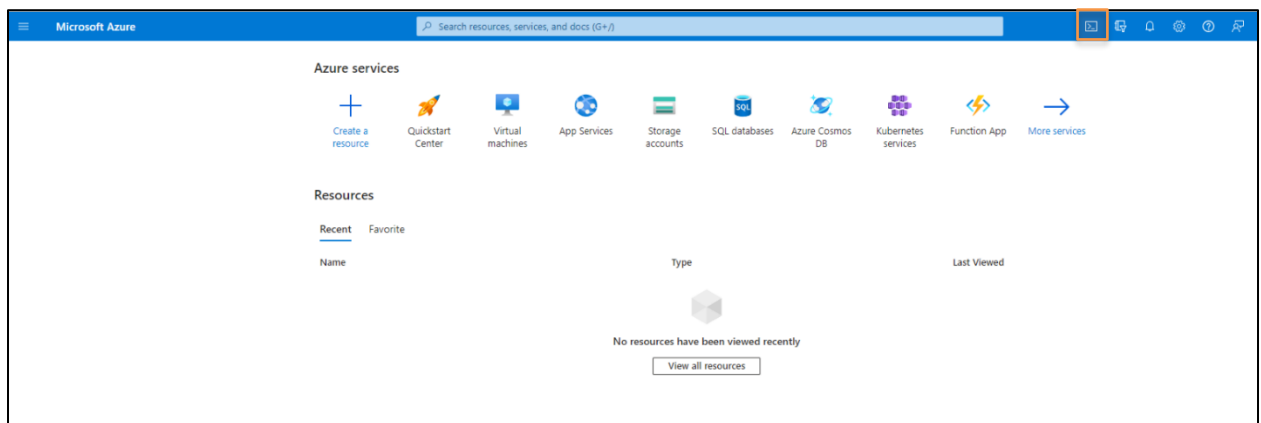
Prerequisites: None

Steps to be followed:

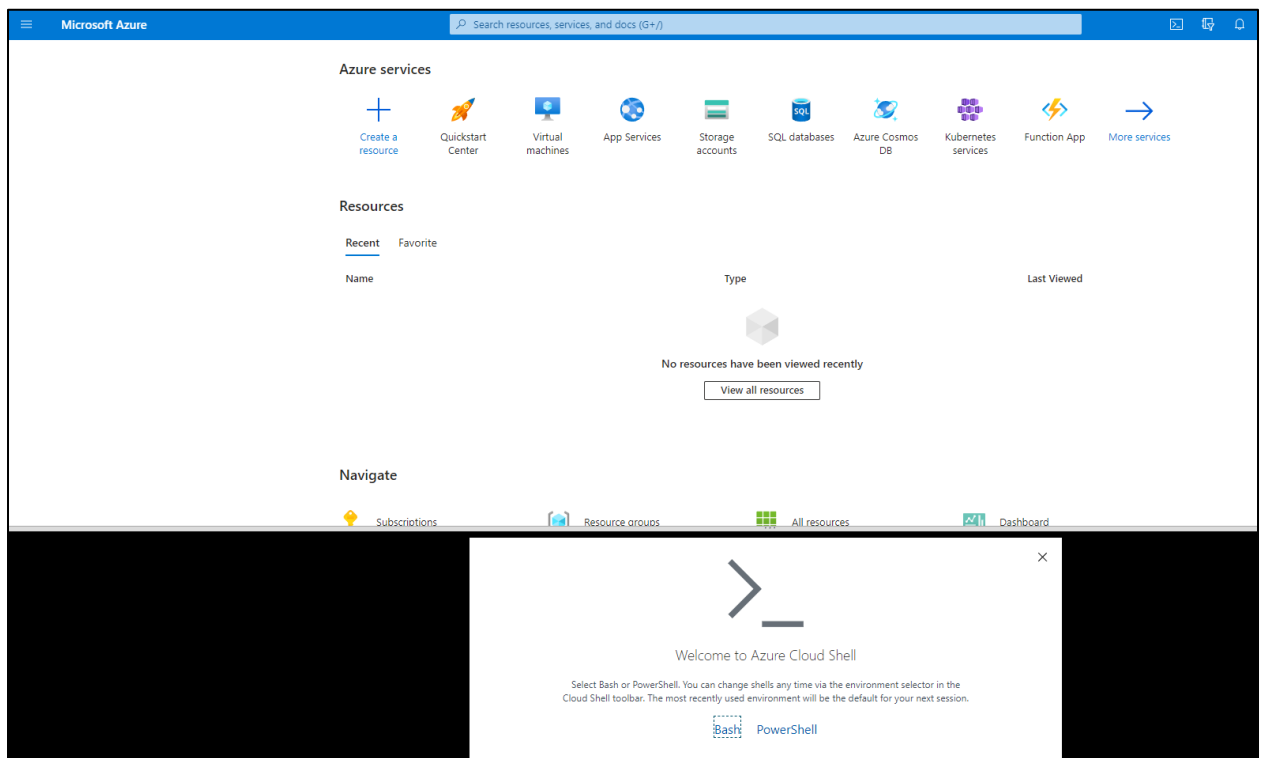
1. Provision the lab environment
2. Configure the local and global virtual network peering
3. Test intersite connectivity

Step 1: Provision the lab environment

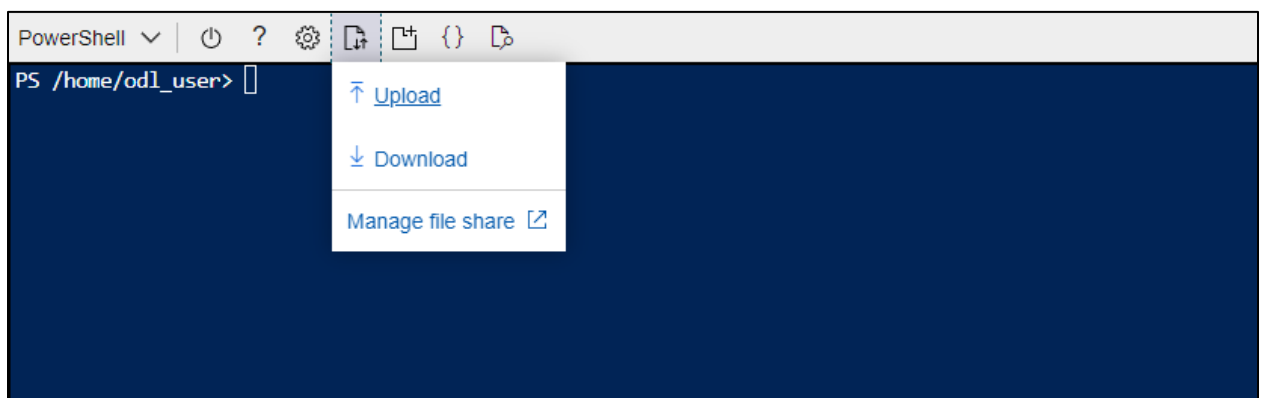
- 1.1 In the Azure portal, access **Azure Cloud Shell** by clicking on its icon at the top right of the portal



1.2 Select PowerShell



1.3 In the toolbar of the Cloud Shell pane, click the **Upload** icon. From the drop-down menu, click **Upload** and upload the files **az305-vnet vm-loop-template.json** and **az305-vnet vm-loop-parameters.json** into the Cloud Shell home directory.



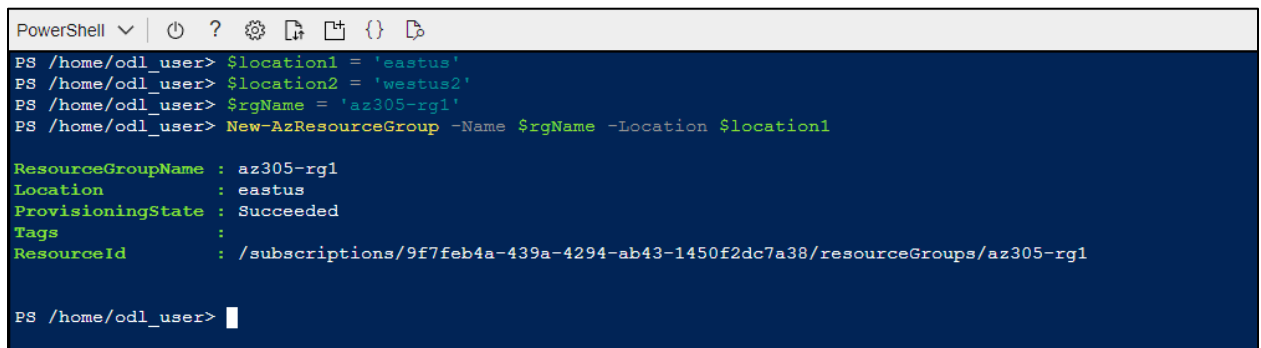
- 1.4 From the Cloud Shell pane, run the following commands to create the resource group that will be hosting the lab environment:

\$location1 = 'eastus'

\$location2 = 'westus2'

\$rgName = 'az305-rg1'

New-AzResourceGroup -Name \$rgName -Location \$location1



```
PowerShell
PS /home/odl_user> $location1 = 'eastus'
PS /home/odl_user> $location2 = 'westus2'
PS /home/odl_user> $rgName = 'az305-rg1'
PS /home/odl_user> New-AzResourceGroup -Name $rgName -Location $location1

ResourceGroupName : az305-rg1
Location           : eastus
ProvisioningState   : Succeeded
Tags               :
ResourceId          : /subscriptions/9f7feb4a-439a-4294-ab43-1450f2dc7a38/resourceGroups/az305-rg1

PS /home/odl_user>
```

- 1.5 From the Cloud Shell pane, run the following commands to create three virtual networks and deploy virtual machines into them by using the template and parameter files:

New-AzResourceGroupDeployment `

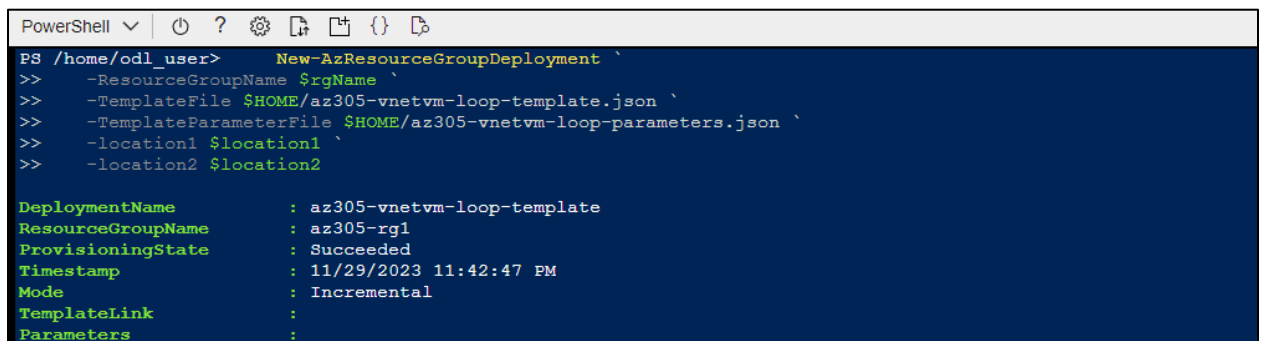
-ResourceGroupName \$rgName `

-TemplateFile \$HOME/az305-vnetvm-loop-template.json `

-TemplateParameterFile \$HOME/az305-vnetvm-loop-parameters.json `

-location1 \$location1 `

-location2 \$location2



```
PowerShell
PS /home/odl_user> New-AzResourceGroupDeployment `
>> -ResourceGroupName $rgName `
>> -TemplateFile $HOME/az305-vnetvm-loop-template.json `
>> -TemplateParameterFile $HOME/az305-vnetvm-loop-parameters.json `
>> -location1 $location1 `
>> -location2 $location2

DeploymentName      : az305-vnetvm-loop-template
ResourceGroupName   : az305-rg1
ProvisioningState    : Succeeded
Timestamp           : 11/29/2023 11:42:47 PM
Mode                : Incremental
TemplateLink         :
Parameters           :
```

The screenshot shows the Microsoft Azure portal interface. At the top, the search bar contains the text 'Virtual networks'. Below the search bar, the results are categorized into 'All', 'Services (34)', and 'Marketplace (6)'. Under the 'All' category, there are links for 'Documentation (99+)', 'Resources (0)', 'Resource Groups (0)', and 'Microsoft Entra ID (0)'. The 'Services' section is expanded, showing a list of services related to 'Virtual networks'. The first service, 'Virtual networks', is highlighted. Other services listed include 'Virtual networks (classic)', 'Virtual network gateways', 'Virtual Network Managers', 'Virtual Network Terminal Access Points', 'Virtual Network Verifiers', and 'Virtual machines'. At the bottom of the search results, there is a section for 'Network managers' with the keyword 'virtual network'.

2.2 In the list of **Virtual networks**, click **az305-vnet0**

Microsoft Azure

Search resources, services, and docs (G+/)

odl_user_1159969@sim... TAXICAB.COM (SIMPLILEARNHO...)

Home >

Virtual networks

taxicab.com (simplilearnhol16.onmicrosoft.com)

+ Create Manage view Refresh Export to CSV Open query Assign tags

Filter for any field... Subscription equals all Resource group equals all Location equals all Add filter

Showing 1 to 4 of 4 records. No grouping List view

Name ↑↓	Resource group ↑↓	Location ↑↓	Subscription ↑↓
az305-vnet0	az305-rg1	East US	Simplilearn HOL 16
az305-vnet1	az305-rg1	East US	Simplilearn HOL 16
az305-vnet2	az305-rg1	West US 2	Simplilearn HOL 16
vnet	ODL-azure-1159969	East US	Simplilearn HOL 16

2.3 On the **az305-vnet0** virtual network blade, click **Peerings** in the **Settings** section, and click **+ Add**

Microsoft Azure

Search resources, services, and docs (G+/)

odl_user_1159969@sim... TAXICAB.COM (SIMPLILEARNHO...)

Home > Virtual networks > az305-vnet0

az305-vnet0 | Peerings

Virtual network

Search + Add Refresh Sync

Filter by name... Peering status == all

Name ↑↓	Peering status ↑↓	Peer ↑↓	Gateway transit ↑↓
Add a peering to get started			

Settings

- Address space
- Connected devices
- Subnets
- Bastion
- DDoS protection
- Firewall
- Microsoft Defender for Cloud
- Network manager
- DNS servers
- Peerings**
- Service endpoints
- Private endpoints
- Properties
- Locks

2.4 Add peering with the following settings and click **Add**:

Microsoft Azure

Search resources, services, and docs (G+)

odl_user_1159969@sim...
TAXICAB.COM (SIMPLILEARNHO...)

Home > Virtual networks > az305-vnet0 | Peerings >

Add peering

az305-vnet0

This virtual network

Peering link name *

az305-vnet0_to_az305-vnet1

☒ Allow 'az305-vnet0' to access 'az305-vnet1'

☐ Allow 'az305-vnet0' to receive forwarded traffic from 'az305-vnet1'

☒ Allow gateway in 'az305-vnet0' to forward traffic to 'az305-vnet1'

☐ Enable 'az305-vnet0' to use 'az305-vnet1's' remote gateway

Remote virtual network

Peering link name *

az305-vnet1_to_az305-vnet0

Virtual network deployment model

☒ Resource manager

☐ Classic

☐ I know my resource ID

Subscription *

Simplilearn HOL 16

Virtual network *

az305-vnet1

☒ Allow 'az305-vnet1' to access 'az305-vnet0'

☐ Allow 'az305-vnet1' to receive forwarded traffic from 'az305-vnet0'

☒ Allow gateway in 'az305-vnet1' to forward traffic to 'az305-vnet0'

☐ Enable 'az305-vnet1' to use 'az305-vnet0's' remote gateway

Add

2.5 On the **az305-vnet0** virtual network blade, click **Peerings** under the **Settings** section and **+ Add**

Home > Virtual networks > az305-vnet0

az305-vnet0 | Peerings ☆ ...

Virtual network

Search

Settings

- Address space
- Connected devices
- Subnets
- Bastion
- DDoS protection
- Firewall
- Microsoft Defender for Cloud
- Network manager
- DNS servers
- Peerings**
- Service endpoints
- Private endpoints
- Properties
- Locks

+ Add Refresh Sync

Filter by name... Peering status == all

Name ↑↓	Peering status ↑↓	Peer ↑↓	Gateway transit ↑↓
az305-vnet0_to_az305-vnet1	Updating	az305-vnet1	Enabled

Microsoft Azure

Search resources, services, and docs (G+/)



Home > Virtual networks > az305-vnet0 | Peerings >

Add peering

az305-vnet0

This virtual network

Peering link name *

az305-vnet0_to_az305-vnet2

☒ Allow 'az305-vnet0' to access 'az305-vnet2'

☐ Allow 'az305-vnet0' to receive forwarded traffic from 'az305-vnet2'

☒ Allow gateway in 'az305-vnet0' to forward traffic to 'az305-vnet2'

☐ Enable 'az305-vnet0' to use 'az305-vnet2's' remote gateway

Remote virtual network

Peering link name *

az305-vnet2_to_az305-vnet0

☒ Resource manager

☐ Classic

☐ I know my resource ID

Subscription *

Simplilearn HOL 16

Virtual network *

az305-vnet2

☒ Allow 'az305-vnet2' to access 'az305-vnet0'

☐ Allow 'az305-vnet2' to receive forwarded traffic from 'az305-vnet0'

☒ Allow gateway in 'az305-vnet2' to forward traffic to 'az305-vnet0'

☐ Enable 'az305-vnet2' to use 'az305-vnet0's' remote gateway

Add

2.7 Navigate to the **Virtual networks** blade and select **az305-vnet1** from the list of available virtual networks

The screenshot shows the Microsoft Azure portal interface for the 'Virtual networks' blade. The breadcrumb trail is 'Home > Virtual networks'. The page title is 'Virtual networks' with a star icon and a dropdown menu. Below the title, there's a search bar and several action buttons: '+ Create', 'Manage view', 'Refresh', 'Export to CSV', 'Open query', and 'Assign tags'. A filter bar shows 'Subscription equals all', 'Resource group equals all', and 'Location equals all'. The main content area displays a table with 4 records. The table has columns: Name, Resource group, Location, and Subscription. The record 'az305-vnet1' is highlighted.

Name	Resource group	Location	Subscription
az305-vnet0	az305-rg1	East US	Simplilearn HOL 16
az305-vnet1	az305-rg1	East US	Simplilearn HOL 16
az305-vnet2	az305-rg1	West US 2	Simplilearn HOL 16
vnet	ODL-azure-1159969	East US	Simplilearn HOL 16

2.8 On the **az305-vnet1** virtual network blade, click on **Peerings** in the **Settings** section and **+ Add**

The screenshot shows the 'az305-vnet1 | Peering' blade in the Microsoft Azure portal. The breadcrumb trail is 'Home > Virtual networks > az305-vnet1'. The page title is 'az305-vnet1 | Peering'. On the left, there's a 'Settings' sidebar with various options: Address space, Connected devices, Subnets, Bastion, DDoS protection, Firewall, Microsoft Defender for Cloud, Network manager, DNS servers, Peering (selected), Service endpoints, Private endpoints, Properties, and Locks. The main content area shows a table with 1 record. The table has columns: Name, Peering status, Peer, and Gateway transit. The record 'az305-vnet1_to_az305-vnet0' is shown with a status of 'Connected' and a peer of 'az305-vnet0'.

Name	Peering status	Peer	Gateway transit
az305-vnet1_to_az305-vnet0	Connected	az305-vnet0	Enabled

2.9 Add peering with the following settings (leave others with their default values) and click **Add**:

[Home](#) > [Virtual networks](#) > [az305-vnet1 | Peerings](#) >

Add peering

az305-vnet1

This virtual network

Peering link name *

☒ Allow 'az305-vnet1' to access 'az305-vnet2'
 ☐ Allow 'az305-vnet1' to receive forwarded traffic from 'az305-vnet2'
 ☒ Allow gateway in 'az305-vnet1' to forward traffic to 'az305-vnet2'
 ☐ Enable 'az305-vnet1' to use 'az305-vnet2's' remote gateway

Remote virtual network

Peering link name *

Virtual network deployment model
 ☒ Resource manager
 ☐ Classic
 ☐ I know my resource ID

Subscription *

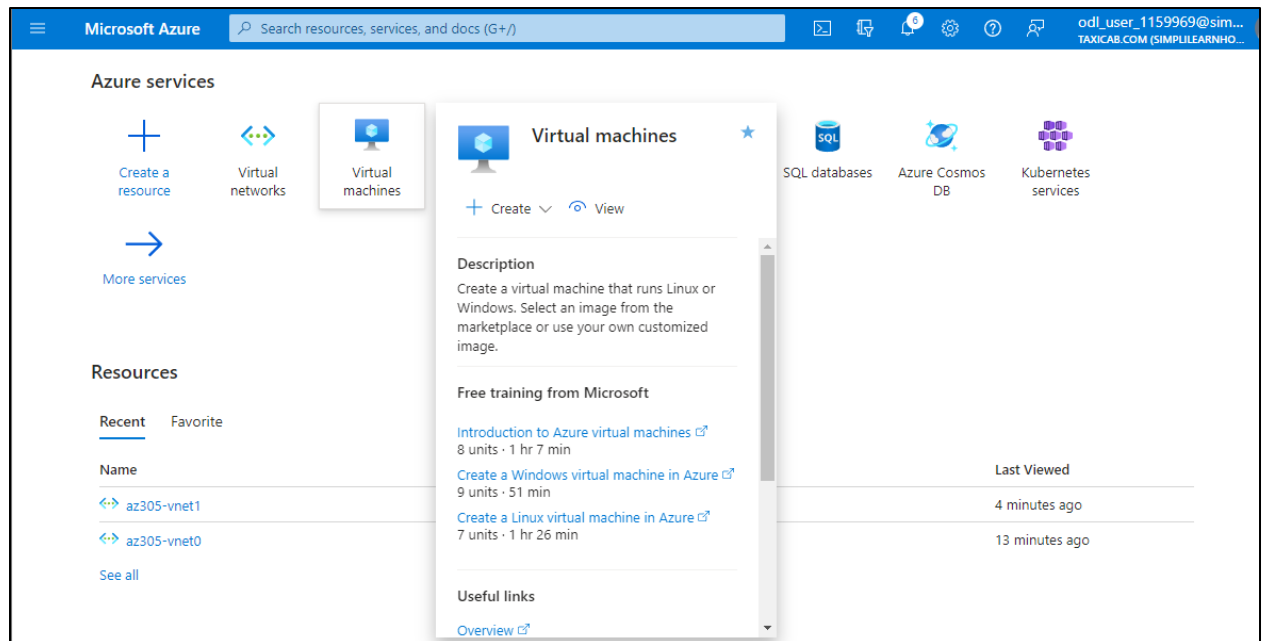
Virtual network *

☒ Allow 'az305-vnet2' to access 'az305-vnet1'
 ☐ Allow 'az305-vnet2' to receive forwarded traffic from 'az305-vnet1'
 ☒ Allow gateway in 'az305-vnet2' to forward traffic to 'az305-vnet1'
 ☐ Enable 'az305-vnet2' to use 'az305-vnet1's' remote gateway

Add

Step 3: Test the Intersite connectivity

3.1 In the Azure portal, search for and select **Virtual machines**



3.2 From the list of virtual machines, click **az305-vm0**

Home > Virtual machines ✎ ...

taxicab.com (simplilearnhol16.onmicrosoft.com)

+ Create ▾ Switch to classic ⌚ Reservations ▾ ⚙️ Manage view ▾ ↻ Refresh ⬇️ Export to CSV 🔗 Open query | 🏷️ Assign tags ▶ Start 🔄 Restart ☐ Stop ...

Filter for any field... Subscription equals all Type equals all Resource group equals all Location equals all Add filter

Showing 1 to 4 of 4 records. No grouping List view

☐ Name ↑↓	Type ↑↓	Subscription ↑↓	Resource group ↑↓	Location ↑↓	Status ↑↓	Operating system ↑↓	Size ↑↓
☐ az305-vm0	Virtual machine	Simplilearn HOL 16	az305-rg1	East US	Running	Windows	Standard_D2s_v3
☐ az305-vm1	Virtual machine	Simplilearn HOL 16	az305-rg1	East US	Running	Windows	Standard_D2s_v3
☐ az305-vm2	Virtual machine	Simplilearn HOL 16	az305-rg1	West US 2	Running	Windows	Standard_D2s_v3
☐ VM-1159969	Virtual machine	Simplilearn HOL 16	ODL-azure-1159969	East US	Running	Windows	Standard_B4ms

3.3 On the **az305-vm0** blade, click **Connect** in the drop-down menu and select the **Connect** blade

Home > Virtual machines > az305-vm0

Virtual machine

Search

Connect Start Restart Stop Hibernate (preview) Capture Delete Refresh Open in mobile

Overview

- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems

Connect

- Connect
- Bastion
- Windows Admin Center

Networking

- Network settings
- Load balancing
- Application security groups
- Network manager

Connect

- Connect via Bastion
- resource group [\(move\)](#)
- [az305-rg1](#)
- Status
- Running
- Location
- East US
- Subscription [\(move\)](#)
- [Simplilearn HOL 16](#)
- Subscription ID
- 9f7feb4a-439a-4294-ab43-1450f2dc7a38
- Tags [\(edit\)](#)
- [Add tags](#)

Operating system

Windows (Windows Server 2019 Datacenter)

Size

Standard D2s v3 (2 vcpus, 8 GiB memory)

Public IP address

[20.124.133.125](#)

Virtual network/subnet

[az305-vnet0/subnet0](#)

DNS name

[Not configured](#)

Health state

-

JSON View

Properties Monitoring Capabilities (8) Recommendations Tutorials

Virtual machine Networking

3.4 Click on the **Select** button

Home > Virtual machines > az305-vm0

az305-vm0 | Connect

Virtual machine

Search

Refresh Troubleshooting More Options Feedback

Connecting using
Public IP address | 20.124.133.125

Admin username
Student

Port (change)
3389 [Check access](#)

Just-in-time policy
Unsupported by plan

Most common

Local machine

Native RDP

Connect via native RDP without any additional software needed. Recommended for testing only.

Public IP address (20.124.133.125)

Select

More ways to connect (4)

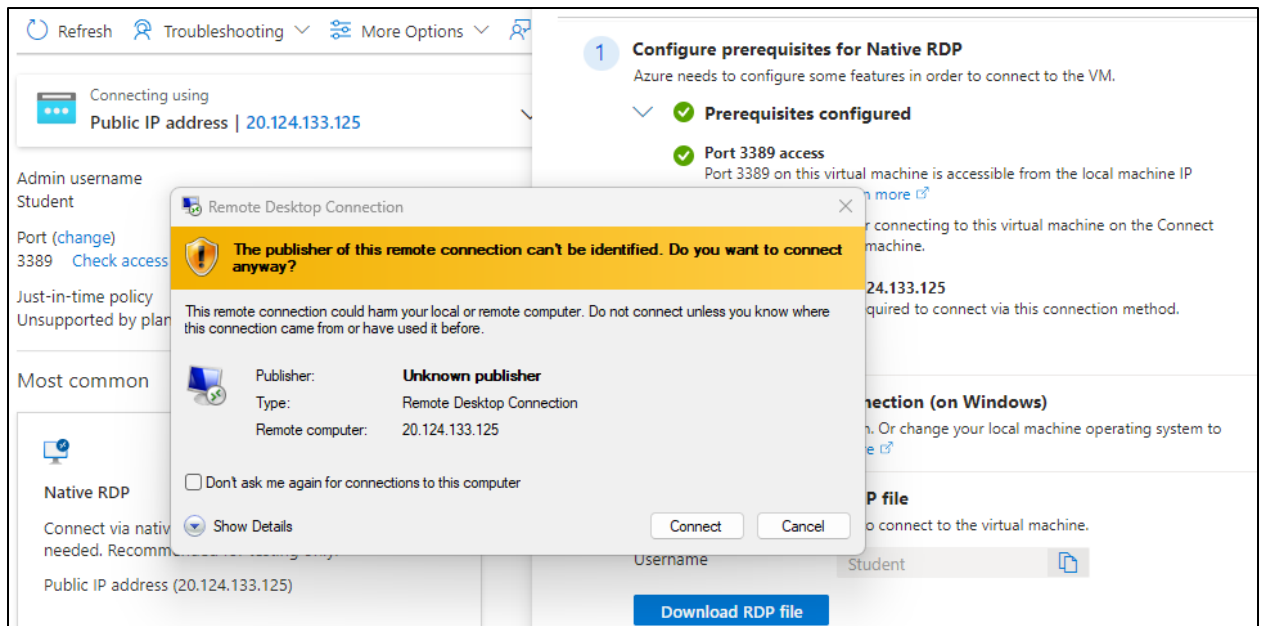
- Overview
- Activity log
- Access control (IAM)
- Tags
- Diagnose and solve problems
- Connect
 - Connect
 - Bastion
 - Windows Admin Center
- Networking
 - Network settings
 - Load balancing
 - Application security groups
 - Network manager
- Settings
 - Disks
 - Extensions + applications
 - Configuration
 - Advisor recommendations

3.5 In the **Native RDP** window, click on **Download RDP file**

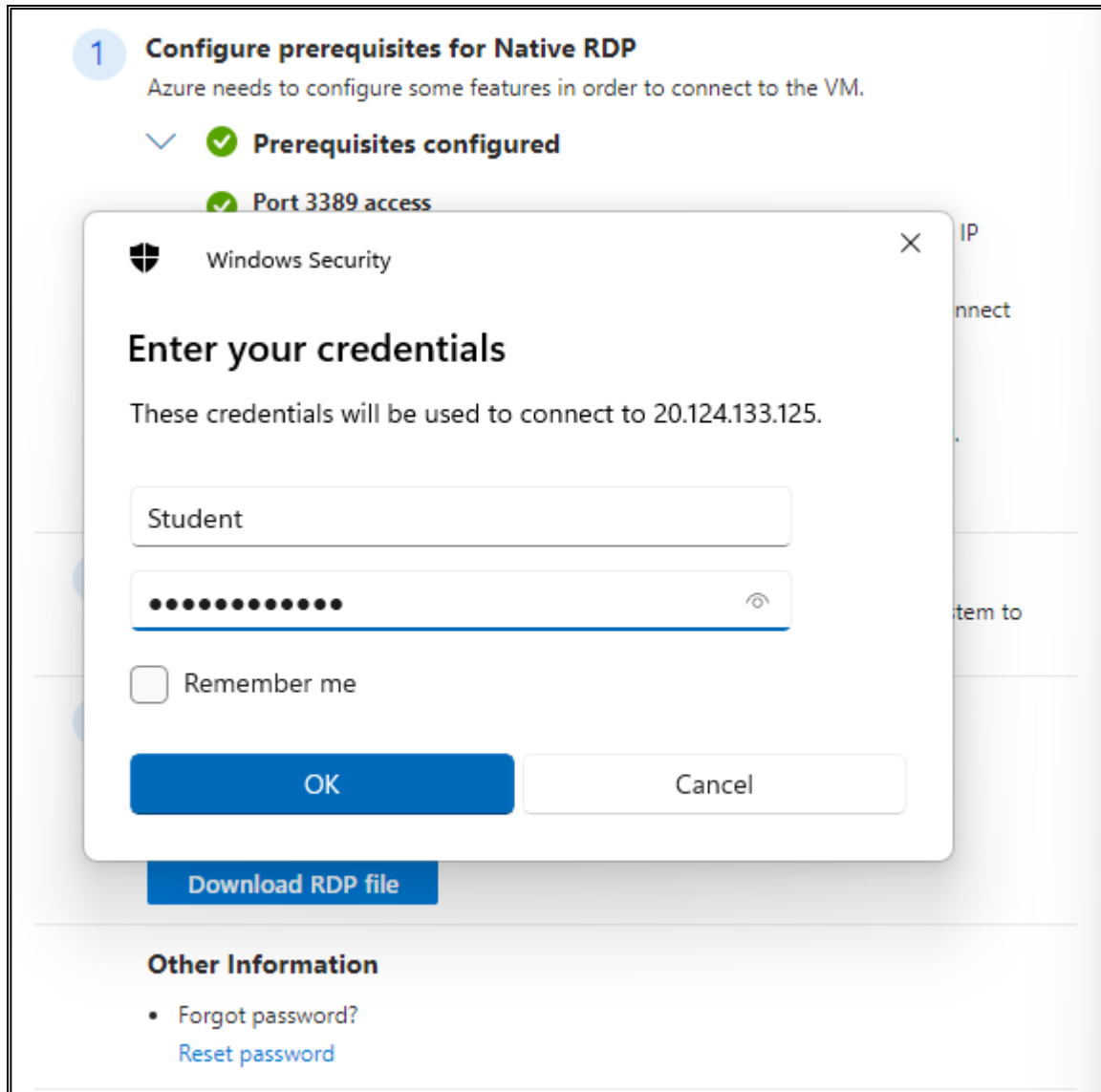
The screenshot shows the Azure portal interface for a virtual machine named 'az305-vm0'. The left sidebar contains navigation links for Overview, Activity log, Access control (IAM), Tags, Diagnose and solve problems, Connect, Bastion, Windows Admin Center, Networking, Network settings, Load balancing, Application security groups, Network manager, Settings, Disks, and Extensions + applications. The 'Connect' section is active, showing connection details: Admin username 'Student', Port '3389', and Just-in-time policy 'Unsupported by plan'. Under 'Most common', the 'Native RDP' option is selected, with a description: 'Connect via native RDP without any additional software needed. Recommended for testing only.' and the Public IP address '20.124.133.125'. A 'Select' button is visible. On the right, the 'Native RDP' configuration window is open, showing three steps: 1. Configure prerequisites for Native RDP (Prerequisites configured, Port 3389 access, Public IP address), 2. Open Remote Desktop Connection (on Windows), and 3. Download and open the RDP file. The 'Download RDP file' button is highlighted. Other information includes 'Forgot password?' and 'Reset password' links.

This screenshot shows the Microsoft Azure portal interface. The top navigation bar includes the Microsoft Azure logo, a search bar, and a 'Recent download history' section. The main content area shows the 'Native RDP' connection window for virtual machine 'az305-vm0'. The 'Download RDP file' button is highlighted, and the file 'az305-vm0.rdp' (96 B) is shown as downloaded. The 'Recent download history' section on the right shows the file 'az305-vm0.rdp' (96 B) with a status of 'Done'.

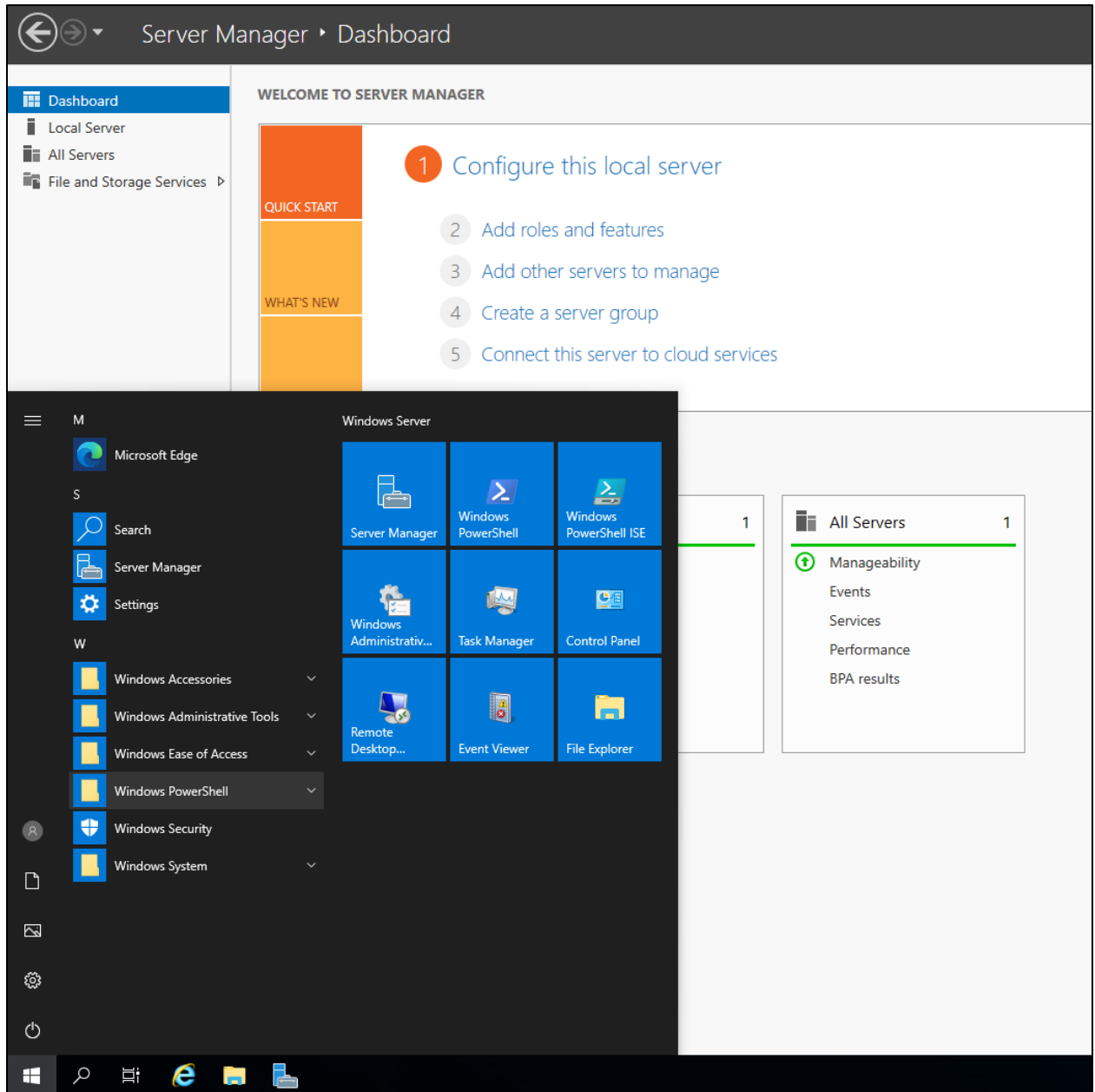
3.6 In the **Remote Desktop Connection** prompt, click **Connect**



3.7 When prompted, sign in by using **Student** as the username and **Pa55w.rd123** as the password. Click on **OK**.

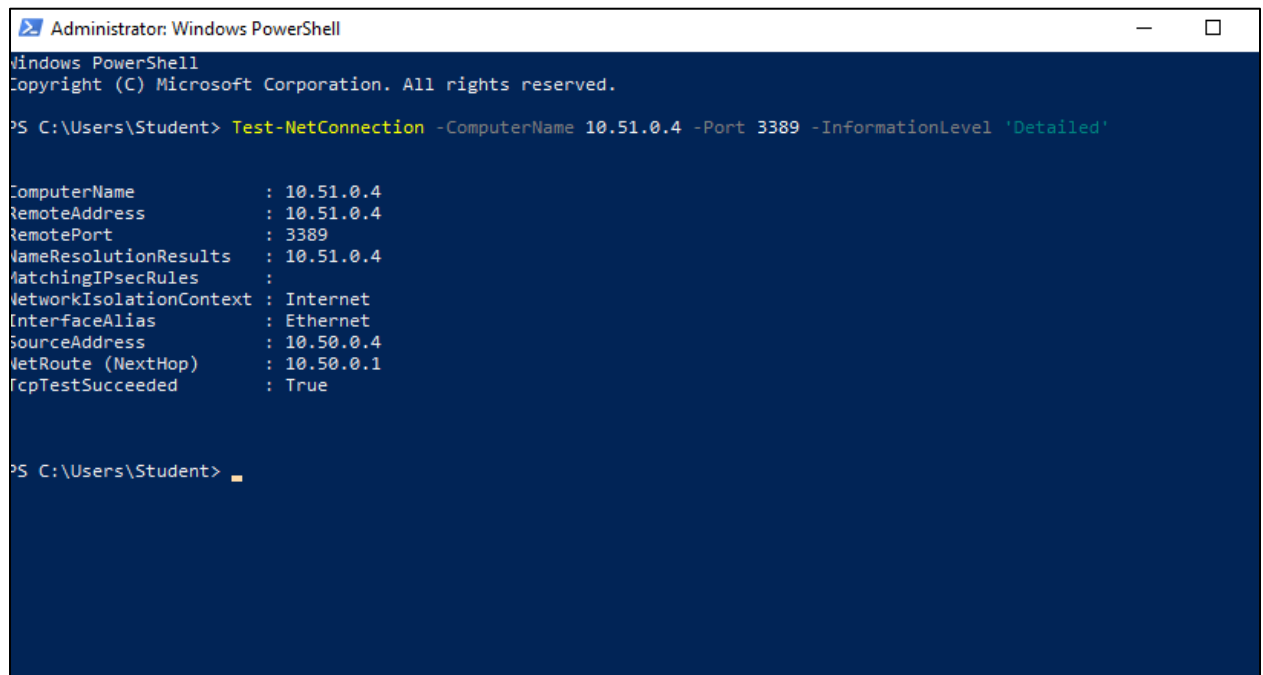


3.8 Within the Remote Desktop session of **az305-vm0**, right-click the **Start** button. In the menu, click on **Windows PowerShell**.



3.9 In the **Windows PowerShell** console window, run the following command to test connectivity to **az305-vm1** (which has a private IP address of **10.51.0.4**) over TCP port 3389:

Test-NetConnection -ComputerName 10.51.0.4 -Port 3389 -InformationLevel 'Detailed'



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Student> Test-NetConnection -ComputerName 10.51.0.4 -Port 3389 -InformationLevel 'Detailed'

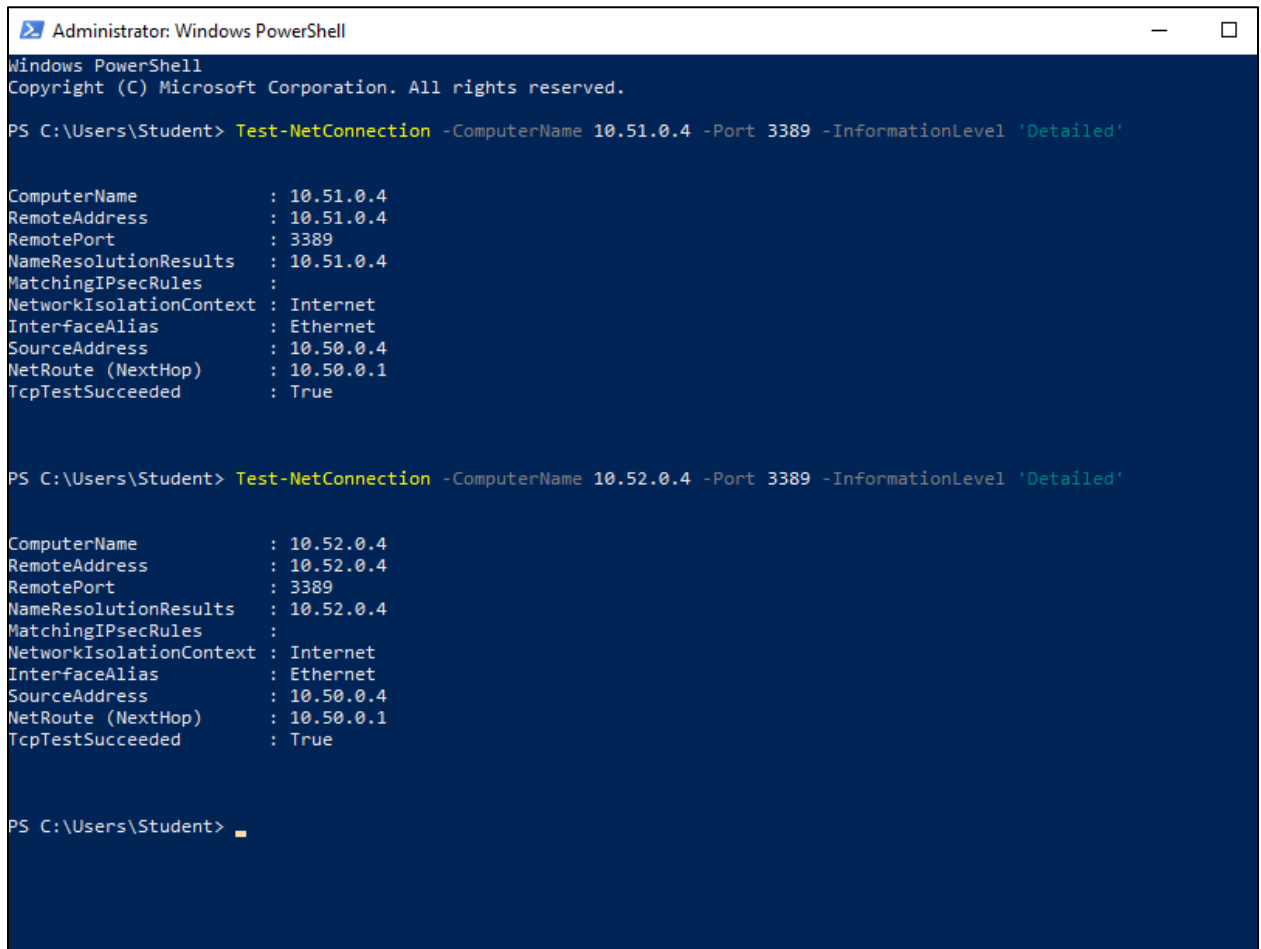
ComputerName           : 10.51.0.4
RemoteAddress          : 10.51.0.4
RemotePort             : 3389
NameResolutionResults  : 10.51.0.4
MatchingIPsecRules     :
NetworkIsolationContext : Internet
InterfaceAlias         : Ethernet
SourceAddress          : 10.50.0.4
NetRoute (NextHop)     : 10.50.0.1
TcpTestSucceeded       : True

PS C:\Users\Student> 
```

Examine the output of the command and verify that the connection was successful.

3.10 In the **Windows PowerShell** console window, run the following command to test connectivity to **az305-vm2** (which has a private IP address of **10.52.0.4**):

Test-NetConnection -ComputerName 10.52.0.4 -Port 3389 -InformationLevel 'Detailed'



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Student> Test-NetConnection -ComputerName 10.51.0.4 -Port 3389 -InformationLevel 'Detailed'

ComputerName           : 10.51.0.4
RemoteAddress           : 10.51.0.4
RemotePort              : 3389
NameResolutionResults   : 10.51.0.4
MatchingIPsecRules      :
NetworkIsolationContext : Internet
InterfaceAlias          : Ethernet
SourceAddress           : 10.50.0.4
NetRoute (NextHop)      : 10.50.0.1
TcpTestSucceeded        : True

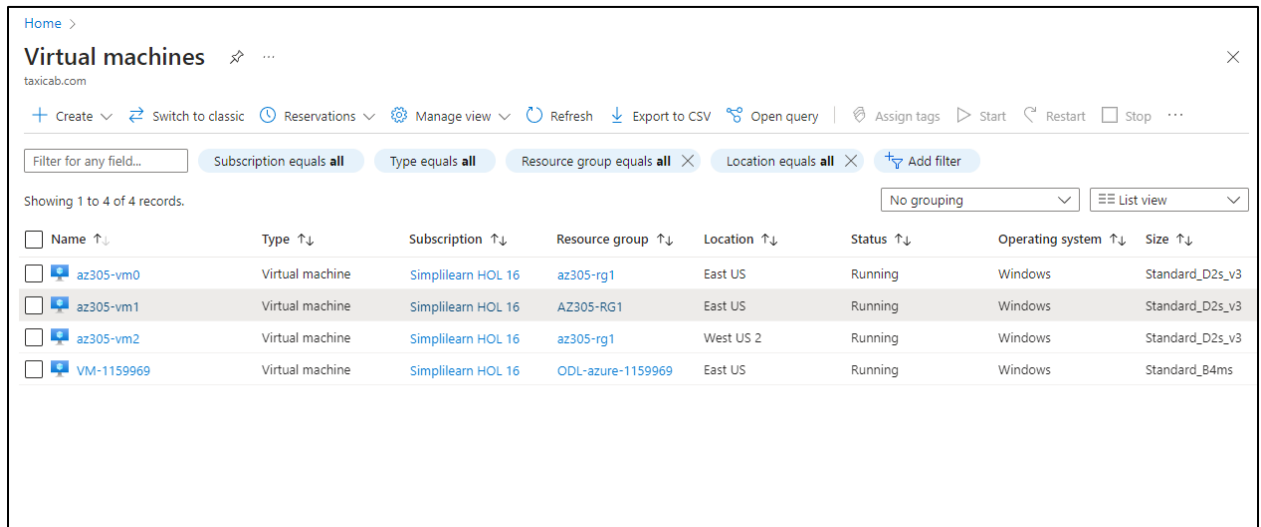
PS C:\Users\Student> Test-NetConnection -ComputerName 10.52.0.4 -Port 3389 -InformationLevel 'Detailed'

ComputerName           : 10.52.0.4
RemoteAddress           : 10.52.0.4
RemotePort              : 3389
NameResolutionResults   : 10.52.0.4
MatchingIPsecRules      :
NetworkIsolationContext : Internet
InterfaceAlias          : Ethernet
SourceAddress           : 10.50.0.4
NetRoute (NextHop)      : 10.50.0.1
TcpTestSucceeded        : True

PS C:\Users\Student> 
```

3.11 Switch to the Azure portal on your lab computer and navigate to the **Virtual machines** blade

3.12 In the list of virtual machines, click **az305-vm1**



Home > Virtual machines taxicab.com

+ Create Switch to classic Reservations Manage view Refresh Export to CSV Open query Assign tags Start Restart Stop

Filter for any field... Subscription equals all Type equals all Resource group equals all Location equals all Add filter

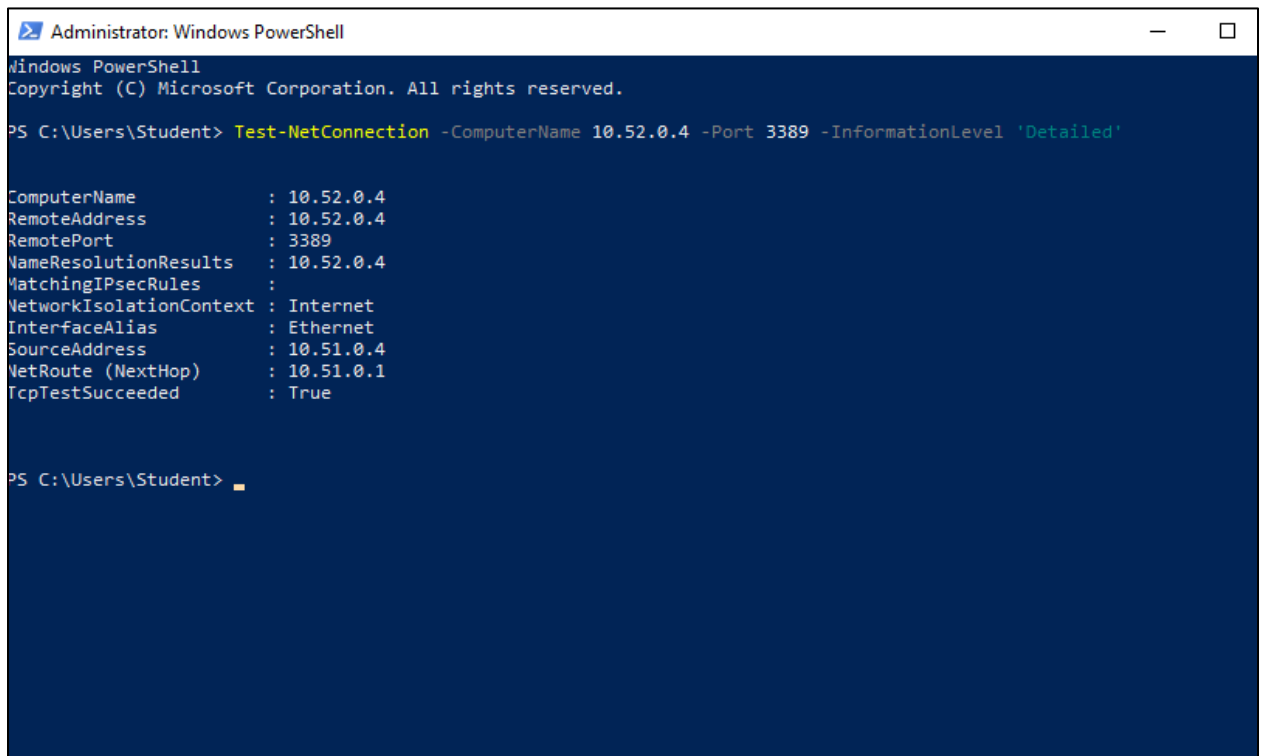
Showing 1 to 4 of 4 records. No grouping List view

Name	Type	Subscription	Resource group	Location	Status	Operating system	Size
az305-vm0	Virtual machine	Simplilearn HOL 16	az305-rg1	East US	Running	Windows	Standard_D2s_v3
az305-vm1	Virtual machine	Simplilearn HOL 16	AZ305-RG1	East US	Running	Windows	Standard_D2s_v3
az305-vm2	Virtual machine	Simplilearn HOL 16	az305-rg1	West US 2	Running	Windows	Standard_D2s_v3
VM-1159969	Virtual machine	Simplilearn HOL 16	ODL-azure-1159969	East US	Running	Windows	Standard_B4ms

Note: Perform steps 3.3 to 3.8 to connect to the virtual machine through RDP access.

- 3.13 In the **Windows PowerShell** console window, run the following command to test connectivity to **az305-vm2** (which has a private IP address of **10.52.0.4**) over TCP port 3389:

Test-NetConnection -ComputerName 10.52.0.4 -Port 3389 -InformationLevel 'Detailed'



```
Administrator: Windows PowerShell
Windows PowerShell
Copyright (C) Microsoft Corporation. All rights reserved.

PS C:\Users\Student> Test-NetConnection -ComputerName 10.52.0.4 -Port 3389 -InformationLevel 'Detailed'

ComputerName           : 10.52.0.4
RemoteAddress          : 10.52.0.4
RemotePort             : 3389
NameResolutionResults  : 10.52.0.4
MatchingIPsecRules     :
NetworkIsolationContext : Internet
InterfaceAlias         : Ethernet
SourceAddress          : 10.51.0.4
NetRoute (NextHop)     : 10.51.0.1
TcpTestSucceeded       : True

PS C:\Users\Student>
```

Examine the output of the command and verify that the connection was successful.

With this, you have successfully implemented a lab environment with different networks having full connectivity and verified its functionality.